# LAPORAN PRAKTIKUM

# KONFIGURASI DNS

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## KONFIGURASI DNS

1. Sebelum melakukan instalasi DNS Server, terlebih dahulu lakukan update dan upgrade package repository agar sistem menggunakan versi paket terbaru dan menghindari error saat instalasi. Jalankan perintah berikut

```
aisasilvana@aisasilvana:~$ sudo apt update && sudo apt upgrade -y
Ign:1 http://id.archive.ubuntu.com/ubuntu noble InRelease
Ign:2 http://id.archive.ubuntu.com/ubuntu noble-updates InRelease
Ign:3 http://id.archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:1 http://id.archive.ubuntu.com/ubuntu noble InRelease
Ign:2 http://id.archive.ubuntu.com/ubuntu noble-updates InRelease
Ign:3 http://id.archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:1 http://id.archive.ubuntu.com/ubuntu noble InRelease
Ign:2 http://id.archive.ubuntu.com/ubuntu noble-updates InRelease
Ign:3 http://id.archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Err:1 http://id.archive.ubuntu.com/ubuntu noble InRelease
  Temporary failure resolving 'id.archive.ubuntu.com'
Err:2 http://id.archive.ubuntu.com/ubuntu noble-updates InRelease
  Temporary failure resolving 'id.archive.ubuntu.com'
Err:4 http://security.ubuntu.com/ubuntu noble-security InRelease
  Temporary failure resolving 'security.ubuntu.com'
Err:3 http://id.archive.ubuntu.com/ubuntu noble-backports InRelease
  Temporary failure resolving 'id.archive.ubuntu.com'
Reading package lists... 24%
```

2. Setelah proses update dan upgrade selesai, install DNS Server menggunakan paket Bind9 dengan perintah:

```
sudo apt install bind9 bind9utils bind9-doc -y
```

```
aisasilvana@aisasilvana:~$ sudo apt install bind9 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  bind-doc
The following NEW packages will be installed:
  bind9
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 258 kB of archives.
After this operation, 963 kB of additional disk space will be used.
Get:1 http://id.archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9 amd64 1:9.18.39-0ubunt
Fetched 258 kB in 1s (201 kB/s)
Selecting previously unselected package bind9.
(Sedang membaca basis data ...  126615 berkas atau direktori telah terpasang.)
Preparing to unpack .../bind9_1%3a9.18.39-0ubuntu0.24.04.3_amd64.deb ...
Unpacking bind9 (1:9.18.39-0ubuntu0.24.04.3) ...
Sedang menata bind9 (1:9.18.39-0ubuntu0.24.04.3) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group `bind' (GID 110) ...
info: Selecting UID from range 100 to 999 ...

info: Adding system user `bind' (UID 109) ...
info: Adding new user `bind' (UID 109) with group `bind' ...
info: Not creating home directory `/var/cache/bind'.
wrote key file "/etc/bind/rndc.key"
named-resolvconf.service is a disabled or a static unit, not starting it.
Created symlink /etc/systemd/system/bind9.service → /usr/lib/systemd/system/named.service.
Created symlink /etc/systemd/system/multi-user.target.wants/named.service → /usr/lib/systemd/sy

Progress: [ 60%] [#################################################################################
```

3. Setelah berhasil diinstal jalankan ulang dan cek statusnya

```
aisasilvana@aisasilvana:~$ sudo systemctl start named
aisasilvana@aisasilvana:~$ sudo systemctl enable named
Synchronizing state of named.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable named
aisasilvana@aisasilvana:~$ sudo systemctl status named
● named.service - BIND Domain Name Server
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-04-22 02:30:43 UTC; 2min 58s ago
       Docs: man:named(8)
   Main PID: 1909 (named)
     Status: "running"
      Tasks: 8 (limit: 4544)
     Memory: 24.3M (peak: 24.8M)
        CPU: 655ms
     CGroup: /system.slice/named.service
             └─1909 /usr/sbin/named -f -u bind

Apr 22 02:33:34 aisasilvana named[1909]: network unreachable resolving 'org/NS/IN': 2001:500:1::53#53
Apr 22 02:33:34 aisasilvana named[1909]: network unreachable resolving 'org/NS/IN': 2001:500:9f::42#53
Apr 22 02:33:34 aisasilvana named[1909]: network unreachable resolving 'org/NS/IN': 2001:500:2d::d#53
Apr 22 02:33:34 aisasilvana named[1909]: network unreachable resolving 'org/NS/IN': 2001:7fd::1#53
Apr 22 02:33:34 aisasilvana named[1909]: network unreachable resolving 'org/NS/IN': 2001:503:c27::2:30#53
Apr 22 02:33:36 aisasilvana named[1909]: network unreachable resolving './NS/IN': 2001:500:12::d0d#53
Apr 22 02:33:36 aisasilvana named[1909]: network unreachable resolving 'org/NS/IN': 2001:500:12::d0d#53
Apr 22 02:33:37 aisasilvana named[1909]: network unreachable resolving './NS/IN': 2001:7fe::53#53
Apr 22 02:33:37 aisasilvana named[1909]: network unreachable resolving 'org/NS/IN': 2001:7fe::53#53
Apr 22 02:33:41 aisasilvana named[1909]: resolver priming query complete: timed out
aisasilvana@aisasilvana:~$
```

4. Setelah instalasi Bind9 selesai, langkah berikutnya adalah mengatur file named.conf.options untuk menentukan DNS forwarder dan izin akses query.

```
  GNU nano 7.2                                        /etc/bind/named.conf.options *
options {
    directory "/var/cache/bind";

    forwarders {
        8.8.8.8;
        8.8.4.4;
    };

    allow-query { any; };
    listen-on { 192.168.100.1; 127.0.0.1; };

    dnssec-validation no;
};
```

5. Selanjutnya lakukan konfigurasi zone forward dan reverse lookup pada file named.conf.local.

```
  GNU nano 7.2                                                    /etc/bind/named.conf.local *
//_
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "aisasilvana.local" {
    type master;
    file "/etc/bind/zones/forward.aisasilvana";
    allow-update { none; };
};

zone "100.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/zones/reverse.aisasilvana";
    allow-update { none; };
};
```

6. Kemudian buat Direktori Zone DNS, Karena file zone akan disimpan di folder /etc/bind/zones, maka terlebih dahulu buat direktori tersebut dengan perintah:

```
sudo mkdir /etc/bind/zones
```

```
aisasilvana@aisasilvana:~$ sudo mkdir /etc/bind/zones
\\aisasilvana@aisasilvana:~$
```

7. Setelah direktori zone berhasil dibuat, langkah berikutnya adalah membuat file forward lookup yang digunakan untuk menerjemahkan nama domain menjadi alamat IP.

```
aisasilvana@aisasilvana:~$ sudo nano /etc/bind/zones/forward.aisasilvana
```

Kemudian edit file tersebut Isi konfigurasi seperti berikut:

```
  GNU nano 7.2                                        /etc/bind/zones/forward.aisasilvana *
$TTL    604800
@       IN      SOA     ns1.aisasilvana.local. root.aisasilvana.local. (
                        2026042201 ; Serial
                        3600       ; Refresh
                        1800       ; Retry
                        604800     ; Expire
                        86400 )    ; Negative TTL
;
@       IN      NS      ns1.aisasilvana.local.
ns1     IN      A       192.168.100.1
www     IN      A       192.168.100.1
```

8. Selanjutnya buat file reverse lookup yang digunakan untuk menerjemahkan alamat IP menjadi nama domain.

```
aisasilvana@aisasilvana:~$ sudo nano /etc/bind/zones/reverse.aisasilvana_
```

Kemudian edit file Isi konfigurasi berikut:

```
  GNU nano 7.2                                                    /etc/bind/zones/reverse.aisasilvana *
$TTL    604800
@       IN      SOA     ns1.aisasilvana.local. root.aisasilvana.local. (
                        2026042201 ; Serial
                        3600       ; Refresh
                        1800       ; Retry
                        604800     ; Expire
                        86400 )    ; Negative TTL
;
@       IN      NS      ns1.aisasilvana.local.
1       IN      PTR     ns1.aisasilvana.local.
```

9. Sebelum menjalankan layanan DNS, lakukan pengecekan syntax konfigurasi agar tidak terjadi error. Cek konfigurasi utama, Cek zone forward dan Cek zone reverse

```
aisasilvana@aisasilvana:~$ sudo named-checkconf
aisasilvana@aisasilvana:~$ sudo named-checkzone aisasilvana.local /etc/bind/zones/forward.aisasilvana
zone aisasilvana.local/IN: loaded serial 2026042201
OK
aisasilvana@aisasilvana:~$ sudo named-checkzone 100.168.192.in-addr.arpa /etc/bind/zones/reverse.aisasilvana
zone 100.168.192.in-addr.arpa/IN: loaded serial 2026042201
OK
aisasilvana@aisasilvana:~$
```

10. Kemudian Restart layanan Bind9 agar semua konfigurasi dapat diterapkan

```
aisasilvana@aisasilvana:~$ sudo systemctl restart named
aisasilvana@aisasilvana:~$ sudo systemctl status named
● named.service - BIND Domain Name Server
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-04-22 02:44:59 UTC; 10s ago
       Docs: man:named(8)
   Main PID: 2216 (named)
     Status: "running"
      Tasks: 6 (limit: 4544)
     Memory: 22.9M (peak: 23.3M)
        CPU: 80ms
     CGroup: /system.slice/named.service
             └─2216 /usr/sbin/named -f -u bind

Apr 22 02:45:02 aisasilvana named[2216]: network unreachable resolving './NS/IN': 2001:503:c27::2:30#53
Apr 22 02:45:02 aisasilvana named[2216]: network unreachable resolving './NS/IN': 2801:1b8:10::b#53
Apr 22 02:45:02 aisasilvana named[2216]: network unreachable resolving './NS/IN': 2001:500:9f::42#53
Apr 22 02:45:02 aisasilvana named[2216]: network unreachable resolving './NS/IN': 2001:dc3::35#53
Apr 22 02:45:02 aisasilvana named[2216]: network unreachable resolving './NS/IN': 2001:500:a8::e#53
Apr 22 02:45:02 aisasilvana named[2216]: network unreachable resolving './NS/IN': 2001:500:1::53#53
Apr 22 02:45:02 aisasilvana named[2216]: network unreachable resolving './NS/IN': 2001:500:2::c#53
Apr 22 02:45:02 aisasilvana named[2216]: network unreachable resolving './NS/IN': 2001:500:12::d0d#53
Apr 22 02:45:02 aisasilvana named[2216]: network unreachable resolving './NS/IN': 2001:500:2f::f#53
Apr 22 02:45:02 aisasilvana named[2216]: network unreachable resolving './NS/IN': 2001:500:2d::d#53
aisasilvana@aisasilvana:~$
```

11. Lakukan pengujian DNS menggunakan perintah berikut

```
aisasilvana@aisasilvana:~$ nslookup ns1.aisasilvana.local
Server:         192.168.100.1
Address:        192.168.100.1#53

Name:   ns1.aisasilvana.local
Address: 192.168.100.1

aisasilvana@aisasilvana:~$ dig ns1.aisasilvana.local

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> ns1.aisasilvana.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 18693
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 0ef1babb846d45540100000069e96afbe82efb1e179f9bbf (good)
;; QUESTION SECTION:
;ns1.aisasilvana.local.         IN      A

;; ANSWER SECTION:
ns1.aisasilvana.local.  604800  IN      A       192.168.100.1

;; Query time: 1 msec
;; SERVER: 192.168.100.1#53(192.168.100.1) (UDP)
;; WHEN: Thu Apr 23 00:42:35 UTC 2026
;; MSG SIZE  rcvd: 94

aisasilvana@aisasilvana:~$
```